Vectors and Embeddings in AI

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Vectors, Norms & High-Dimensional Embeddings · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

2 **[Foundational]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Probability Distributions & Maximum Likelihood (MLE) · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

3 **[Foundational]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Loss Functions, Gradients & Optimization · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

4 **[Foundational]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Linear Regression & Ordinary Least Squares · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

5 **[Foundational]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Logistic Regression & Classification Boundaries · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

6 **[Foundational]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Matrices, Linear Layers & Activation Functions · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

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7 **[Foundational]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Dimensionality Reduction, Covariance & PCA · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Conditional Probability, Bayes' Rule & Classifiers · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

9 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Vectors in $\mathbb{R}^n$ & Inner Product Spaces · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

10 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

[Vectors & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

11 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

Practice Problem 2: Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y, 2x, (x+y)e^z \rangle$ and C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = y + 1$, oriented counterclockwise when viewed from above.

Compute $abla \times \mathbf{F}$: $abla \times \mathbf{F} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; \partial_x, \partial_y, \partial_z; -y, 2x, (x+y)e^z| = \langle e^z - 0,\ 0 - e^z,\ 2 - (-1) \rangle = \langle e^z,\ -e^z, 3 \rangle.$ Choose the planar surface S bounded by C on the plane $z = y + 1$. Here $z = g(x, y) = y + 1$. Upward normal vector: $\mathbf{n}\, dS = \langle -g_x,\ -g_y, 1 \rangle\, dA = \langle 0,\ -1, 1 \rangle\, dA$. Dot product.

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12 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

Practice Problem 1: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle e^y, xe^y + (z+1)e^z, (z+2)e^z \rangle$ along the curve $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ for $0 \le t \le 1$.

First check if $\mathbf{F}$ is conservative: $P_y = e^y,\ Q_x = e^y;\quad P_z = 0,\ R_x = 0;\quad Q_z = e^z + (z+1)e^z = (z+2)e^z,\ R_y = 0.$ Notice $R_y = 0$ while $Q_z = (z+2)e^z eq 0$. Thus $\mathbf{F}$ is not conservative and the Fundamental Theorem cannot be used directly. Evaluate directly: $\mathbf{r}'(t) = \langle 1, 2t, 3t^2 \rangle$. $x = t, y = t^2, z = t^3$. $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = e^{t^2}(1) + \left[te^{t^2} + (t^3+1)e^{t^3}\right](2t) + (t^3+2)e^{t^3}(3t^2).$ Regro.

13 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$, find the tangential component of acceleration a_T and the normal component a_N at $t = 1$.

Compute derivatives: $\mathbf{r}'(t) = \langle 1, 2t, 2t^2 \rangle$ $\mathbf{r}''(t) = \langle 0, 2, 4t \rangle$ At $t = 1$:
$\mathbf{r}'(1) = \langle 1, 2, 2 \rangle \Rightarrow |\mathbf{r}'(1)| = \sqrt{1+4+4} = 3$ $\mathbf{r}''(1) = \langle 0, 2, 4 \rangle$ Tangential acceleration a_T:
$a_T = \frac{\mathbf{r}' \cdot \mathbf{r}''}{|\mathbf{r}'|} = \frac{(1)(0)+(2)(2)+(2)(4)}{3} = \frac{0+4+8}{3} = \frac{12}{3} = 4$ Normal acceleration a_N: Compute cross product at $t = 1$.

14 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

Practice Problem 1: Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + \sin(yz),\ y^3 + \cos(xz),\ z^3 + e^{xy} \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

Direct surface integration is prohibitive due to cross-coupling transcendental terms. Compute the divergence: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3 + \sin(yz)) + \frac{\partial}{\partial y}(y^3 + \cos(xz)) + \frac{\partial}{\partial z}(z^3 + e^{xy}) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2).$ Notice that all transcendental cross-terms vanish identically. Apply the Divergence Theorem: $\iint_{\partial E} \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\, dV.$ Convert to spherical coordinates where.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Vectors & Embeddings in AI Practice Exercises** • **Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

Prove the Parallelogram Law for arbitrary vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$: $||\mathbf{u} + \mathbf{v}||^2 + ||\mathbf{u} - \mathbf{v}||^2 = 2||\mathbf{u}||^2 + 2||\mathbf{v}||^2$. What is its geometric significance?

Algebraic Proof: Expand each squared Euclidean norm using the inner product:
$||\mathbf{u} + \mathbf{v}||^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = ||\mathbf{u}||^2 + 2(\mathbf{u} \cdot \mathbf{v}) + ||\mathbf{v}||^2.$ $||\mathbf{u} - \mathbf{v}||^2 = (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = ||\mathbf{u}||^2 - 2(\mathbf{u} \cdot \mathbf{v}) + ||\mathbf{v}||^2.$ Adding the two equations directly cancels the cross terms $2(\mathbf{u} \cdot \mathbf{v})$ and.

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16 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Level 2 · MediumFind the distance from the point $P(3, -1, 4)$ in $\mathbb{R}^3$ to the line passing through the origin along the direction vector $\mathbf{d} = (2; 1; -2)$.

Let $\mathbf{p} = (3; -1; 4)$. The orthogonal distance from P to the line $\text{span}(\mathbf{d})$ is $||\mathbf{p}_\perp|| = ||\mathbf{p} - \text{proj}_\mathbf{d}(\mathbf{p})||$. Step 1: Compute projection of $\mathbf{p}$ onto $\mathbf{d}$: $\mathbf{p} \cdot \mathbf{d} = 3(2) + (-1)(1) + 4(-2) = 6 - 1 - 8 = -3$. $||\mathbf{d}||^2 = 2^2 + 1^2 + (-2)^2 = 4 + 1 + 4 = 9$. .

17 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Find the standard matrix $[T]$ of the linear operator $T : \mathbb{R}^2 \to \mathbb{R}^2$ that first rotates vectors by 90° counterclockwise, and then reflects the result across the line $y = x$.

Step 1: Track images of standard basis vectors: For $\mathbf{e}_1 = (1; 0)$: Rotation by 90° CCW sends $\mathbf{e}_1$ to $(0; 1)$. Reflection across $y = x$ swaps coordinates: $(0; 1) \longrightarrow (1; 0)$. For $\mathbf{e}_2 = (0; 1)$: Rotation by 90° CCW sends $\mathbf{e}_2$ to $(-1; 0)$. Reflection across $y = x$ swaps coordinates: $(-1; 0) \longrightarrow (0; -1)$. Step 2: Assemble standard matrix: The.

18 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Level 3 · Advanced / University RigorLet $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ satisfy $||\mathbf{u}|| = 4$, $||\mathbf{v}|| = 5$, and $||\mathbf{u} + \mathbf{v}|| = 7$. Using the Parallelogram Law, find the exact length of the difference vector $||\mathbf{u} - \mathbf{v}||$, and determine the angle θ between $\mathbf{u}$ and $\mathbf{v}$.

Step 1 · Apply the Parallelogram Law: $||\mathbf{u} + \mathbf{v}||^2 + ||\mathbf{u} - \mathbf{v}||^2 = 2||\mathbf{u}||^2 + 2||\mathbf{v}||^2$. Substitute known values: $7^2 + ||\mathbf{u} - \mathbf{v}||^2 = 2(4^2) + 2(5^2) \Rightarrow 49 + ||\mathbf{u} - \mathbf{v}||^2 = 2(16) + 2(25) = 32 + 50 = 82$. $||\mathbf{u} - \mathbf{v}||^2 = 82 - 49 = 33 \Rightarrow \mathbf{||u - v|| = \sqrt{33}}$. Step 2 · Determine the angle θ: Using the Polarization Identity.

19 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Practice Problem 1: Use Green's Theorem to evaluate $\oint_C (y^3 - \ln(x^2 + 1))\,dx + (x^3 + e^{y^2})\,dy$, where C is the circle $x^2 + y^2 = 4$ oriented counterclockwise.

Identify $P = y^3 - \ln(x^2 + 1)$ and $Q = x^3 + e^{y^2}$. Notice that direct line integration would be intractable due to the transcendental terms. Compute the 2D curl: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3x^2 - 3y^2 = 3(x^2 - y^2)$. Notice that both complicated transcendental terms vanish upon differentiation. Apply Green's Theorem over the disk $D = \{(x, y) \mid x^2 + y^2 \leq 4\}$: $\oint_C P\,dx + Q\,dy = \iint_D 3(x^2 - y^2)\,dA$. In polar coordinates: $x^2 - y^2 = r^2\cos(2\theta)$.

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20 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Practice Problem 2: Evaluate the line integral $\int_C (2y + \cos x)\,dx + (5x + \sin y)\,dy$ along the upper half of the unit circle $x^2 + y^2 = 1$ from $(1, 0)$ to $(-1, 0)$.

The curve C is open. Close the loop by defining L as the line segment along the x-axis from $(-1, 0)$ to $(1, 0)$. The composite curve $\Gamma = C + L$ forms a positively oriented boundary enclosing the upper half-disk D. By Green's Theorem: $\oint_\Gamma P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA = \iint_D (5 - 2)\,dA = 3\,\text{Area}(D) = 3\left(\frac{\pi(1)^2}{2}\right) = \frac{3\pi}{2}$. Along L: $y = 0 \Rightarrow dy = 0$, and x runs from -1 to 1.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector \(\mathbf{u} = [2, 3]^T \) with itself.

a. **\(13 \) ✓**

b. \(15 \)

c. \(11 \)

d. \(10 \)

The dot product with itself is the squared Euclidean norm: \(\mathbf{u} \cdot \mathbf{u} = (2)^2 + (3)^2 = 4 + 9 = 13 \).

22 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the Manhattan (\(L_1 \)) norm and Euclidean (\(L_2 \)) norm of the word vector \(\mathbf{w} = [2, -3]^T \).

a. **\(\|\mathbf{w}\|_1 = 5, \; \|\mathbf{w}\|_2 = \sqrt{13} \)** ✓

b. \(\|\mathbf{w}\|_1 = -1, \; \|\mathbf{w}\|_2 = 13 \)

c. \(\|\mathbf{w}\|_1 = 5, \; \|\mathbf{w}\|_2 = 5 \)

d. \(\|\mathbf{w}\|_1 = \sqrt{13}, \; \|\mathbf{w}\|_2 = 5 \)

\(L_1 \) norm is the sum of absolute values: \(|2| + |-3| = 5 \). \(L_2 \) norm is the square root of sum of squares: \(\sqrt{(2)^2 + (-3)^2} = \sqrt{4 + 9} = \sqrt{13} \).

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23 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Consider a dataset of n data points, where each data point is a vector in $\mathbb{R}^D$. A non-linear dimensionality reduction technique maps these data points to a lower-dimensional space $\mathbb{R}^d$. The mapping function for a single data point $\mathbf{x} \in \mathbb{R}^D$ is given by $f(\mathbf{x}) = \phi(\mathbf{x})$, where $\phi : \mathbb{R}^D \to \mathbb{R}^d$ is a function designed to preserve local neighborhood structures. Suppose the mapping ϕ is defined by an explicit feature map $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$ where the vector $\mathbf{x} \in \mathbb{R}^{10}$ and d is the dimension of the embedded space. If the goal is to embed these points into a space such that pairwise Euclidean distances are preserved as closely as possible under a specific distortion, and it is found that $d = 15$ yields optimal results for this mapping under a certain constraint, determine the effective rank of the feature map's image space.

a. 15

b. 10

c. **1** ✓

d. 25

The feature map is defined as $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$, where $\mathbf{x} \in \mathbb{R}^{10}$. The term $\mathbf{x}^T\mathbf{x}$ is the squared Euclidean norm of $\mathbf{x}$, denoted as $||\mathbf{x}||^2$. This is a scalar value. Thus, the mapping ϕ produces a vector where each component is the same scalar value, $||\mathbf{x}||^2$. Let $k = ||\mathbf{x}||^2$. Then $\phi(\mathbf{x}) = [k, k, \ldots, k]^T$ in $\mathbb{R}^d$. Th.

24 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

A vector $\mathbf{v}$ is transformed into a new vector $\mathbf{v}'$ by a linear transformation represented by the matrix $\mathbf{M}$. We are given that $\mathbf{v} = (2\ 3)$ and $\mathbf{v}' = (7\ 11)$. The transformation matrix is of the form $\mathbf{M} = (a, 1\ 4, b)$. Determine the value of a and b.

a. a = 1, b = 3

b. **a = 2, b = 1** ✓

c. a = 3, b = 2

d. a = 4, b = -1

The transformation is defined by $\mathbf{v}' = \mathbf{M}\mathbf{v}$. Substituting the given vectors and matrix form: $(7\ 11) = (a, 1\ 4, b)(2\ 3)$ Performing the matrix multiplication: $(7\ 11) = (2a + 3\ 8 + 3b)$ Equating the corresponding components, we get a system of two linear equations: 1) $2a + 3 = 7$ 2) $8 + 3b = 11$ Solving the first equation for a: $2a = 7 - 3$ $2a = 4$ $a = 2$ Solving the second equation for b: $3b = 11 - 8$ $3b = 3$.

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25 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Consider two vectors, $\mathbf{u} = [3; 4]$ and $\mathbf{v} = [1; -2]$. What is the vector representing $2\mathbf{u} - \mathbf{v}$?

a. $[5; 6]$

b. $[7; 10]$

c. $[5; 10]$ ✓

d. $[2; 12]$

To find the vector $2\mathbf{u} - \mathbf{v}$, we first compute $2\mathbf{u}$ and then subtract $\mathbf{v}$ from the result.
$2\mathbf{u} = 2 \times [3; 4] = [2 \times 3; 2 \times 4] = [6; 8]$. Now, we subtract $\mathbf{v}$ from $2\mathbf{u}$:
$2\mathbf{u} - \mathbf{v} = [6; 8] - [1; -2] = [6 - 1; 8 - (-2)] = [5; 8 + 2] = [5; 10]$. Therefore, the resulting vector is $[5; 10]$.

26 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

A vector $\mathbf{v}$ is represented in a 2D space. If the components of the vector are $v_x = -3$ and $v_y = 4$, what is the Euclidean (L_2) norm of this vector?

a. **5** ✓

b. 7

c. 25

d. 1

The Euclidean (L_2) norm of a vector $\mathbf{v} = [v_x, v_y]^T$ is calculated as $||\mathbf{v}||_2 = \sqrt{v_x^2 + v_y^2}$. For the given vector $\mathbf{v}$ with components $v_x = -3$ and $v_y = 4$, the norm is:
$||\mathbf{v}||_2 = \sqrt{(-3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

27 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Consider two vectors, $\mathbf{u} = (3; -2)$ and $\mathbf{v} = (-1; 5)$. What is the resultant vector $\mathbf{w} = \mathbf{u} + \mathbf{v}$?

a. $(2; 3)$ ✓

b. $(4; -7)$

c. $(2; -7)$

d. $(-4; 3)$

To find the resultant vector $\mathbf{w} = \mathbf{u} + \mathbf{v}$, we add the corresponding components of the two vectors.
$\mathbf{w} = (3; -2) + (-1; 5) = (3 + (-1); -2 + 5) = (2; 3)$.

28 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Two document embeddings $\mathbf{a} = [1, 2]^T$ and $\mathbf{b} = [2, 1]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{5}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800 ✓

d. 0.500

Dot product $\mathbf{a} \cdot \mathbf{b} = (1)(2) + (2)(1) = 4$. Product of norms $||\mathbf{a}||||\mathbf{b}|| = 5$. Cosine similarity = $\frac{4}{5} = 0.800$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Vectors & Embeddings in AI Practice Exercises

A simple linear regression model has been fitted to a dataset. The model equation is given by $\hat{y} = ax + b$. The regression line passes through the points $(2, 10)$ and $(5, 22)$ in the feature-target space. Determine the value of the intercept b.

a. 4

b. **-2** ✓

c. 2

d. -4

The problem states that the linear regression line $\hat{y} = ax + b$ passes through two points: $(x_1, y_1) = (2, 10)$ and $(x_2, y_2) = (5, 22)$. We can use these two points to form a system of linear equations to solve for the unknown parameters a (the slope) and b (the intercept). Substituting the first point $(2, 10)$ into the equation $\hat{y} = ax + b$ gives: $10 = a(2) + b$ (Equation 1) Substituting the second point $(5, 22)$ into the equation $\hat{y} = ax + b$ gives: $22 = a(5) + b$ (Equation 2) To fi.

30 [Intermediate] • Target: Vectors & Embeddings in AI Practice Exercises

Consider a simple logistic regression model where the decision boundary is defined by the equation $3x + 4y - 12 = 0$. This equation represents a linear boundary in a 2D feature space. What is the y-intercept of this decision boundary?

a. $y = -3$

b. $y = 3$ ✓

c. $y = 4$

d. $y = -4$

The equation of the decision boundary is given by $3x + 4y - 12 = 0$. To find the y-intercept, we set $x = 0$ and solve for y. Substituting $x = 0$ into the equation: $3(0) + 4y - 12 = 0$ $4y - 12 = 0$ $4y = 12$ $y = \frac{12}{4}$ $y = 3$ Therefore, the y-intercept of the decision boundary is 3.

31 [Intermediate] • Target: Vectors & Embeddings in AI Practice Exercises

The sigmoid activation function is $\sigma(z) = \frac{1}{1 + e^{-z}}$. If $\sigma(z_0) = 0.20$, what is the value of the derivative $\sigma'(z_0)$?

a. 0.1600 ✓

b. 0.4000

c. 0.8000

d. 0.0400

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.20$: $\sigma'(z_0) = (0.20)(1 - 0.20) = (0.20)(0.80) = 0.1600$.

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32 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

Consider a dataset of n observations (x_i, y_i) for $i = 1, \ldots, n$, where x_i are scalar predictor variables and y_i are scalar response variables. The ordinary least squares (OLS) estimates for the intercept β_0 and the slope β_1 of the linear model $y = \beta_0 + \beta_1 x + \varepsilon$ are given by $\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Suppose we have a dataset where $n = 5$, $\sum x_i = 15$, $\sum y_i = 30$, $\sum x_i^2 = 55$, $\sum y_i^2 = 210$, and $\sum x_i y_i = 105$. A new data point (x_6, y_6) is added to the dataset. If the new regression line that minimizes the sum of squared errors for the $n + 1 = 6$ points is parallel to the original regression line (i.e., $\hat{\beta}_1^{\text{new}} = \hat{\beta}_1^{\text{old}}$), what is the value of y_6 in terms of x_6?

a. $y_6 = 1.5x_6 + 1.5$ ✓

b. $y_6 = 1.5x_6 + 0.5$

c. $y_6 = 2x_6 + 0$

d. $y_6 = 2x_6 + 1.5$

First, we calculate the means: $\bar{x}^{\text{old}} = \frac{\sum x_i}{n} = \frac{15}{5} = 3$ and $\bar{y}^{\text{old}} = \frac{\sum y_i}{n} = \frac{30}{5} = 6$. Next, we calculate the sums of squares and cross-products for the original dataset: $S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 55 - 5(3^2) = 55 - 45 = 10$ $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y} = 105 - 5(3)(6) = 105 - 90 = 15$ The original slope estimate is.

33 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

Consider a discrete random variable X with probability mass function (PMF) given by $P(X = k; \theta) = \frac{\theta^k e^{-\theta}}{k!}$ for $k = 0, 1, 2, \ldots$, where $\theta > 0$ is a parameter. Suppose we have an i.i.d. sample $x_1, x_2, \ldots, x_n$ from this distribution. We want to find the maximum likelihood estimate (MLE) for θ that maximizes the expected value of θ conditional on the sum of the observations, $S = \sum_{i=1}^{n} x_i$. Specifically, we are interested in the value of θ that maximizes $E[\theta | S = s]$. For a Poisson distribution, it is known that $E[\theta | S = s] = \frac{\theta s}{\theta + s}$ if θ itself were distributed as $Gamma(s + 1, 1)$. However, here θ is a fixed but unknown parameter. This question is about finding a parameter value that optimizes a related quantity. Let's consider a modified objective function for estimation that relates to the Fisher Information. Suppose we are interested in maximizing $L(\theta) = \frac{1}{n} \sum_{i=1}^{n} \log P(x_i; \theta) - \theta$. Find the value of θ that maximizes this modified likelihood function, given a sample where $n = 5$ and $\sum_{i=1}^{5} x_i = 15$.

a. 3 ✓

b. 5

c. 15

d. 2

The given distribution is a Poisson distribution with parameter θ, so $P(X = k; \theta) = \frac{\theta^k e^{-\theta}}{k!}$. The log-likelihood for a single observation x_i is $\log P(x_i; \theta) = x_i \log \theta - \theta - \log(x_i!)$. The log-likelihood for the entire sample is $\log L(\theta; x_1, \ldots, x_n) = \sum_{i=1}^{n} (x_i \log \theta - \theta - \log(x_i!)) = \log \theta \sum_{i=1}^{n} x_i - n\theta - \sum_{i=1}^{n} \log(x_i!)$. Let $S = \sum_{i=1}^{n} x_i$. Then.

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34 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

Consider a sequence of n independent Bernoulli trials, where the probability of success in each trial is p. Let X be the random variable representing the number of successes in these n trials. Thus, $X \sim \text{Binomial}(n, p)$. We are given a dataset of N observed values $\{x_1, x_2, \ldots, x_N\}$ for X, where each x_i is an observed count of successes from n trials. The observed data are: [3, 4, 2, 5, 4, 3, 2, 4, 5, 3]. If we assume that $n = 5$ (the number of trials for each observation is fixed at 5), find the value of p that maximizes the likelihood of observing this dataset. Then, compute the maximum likelihood estimate (MLE) of the probability of success, p, and round your answer to two decimal places.

a. **0.70** ✓

b. 0.75

c. 0.60

d. 0.67

The random variable X follows a Binomial distribution: $X \sim \text{Binomial}(n, p)$. The probability mass function (PMF) of the Binomial distribution is given by $P(X = k|n, p) = \binom{n}{k} p^k (1-p)^{n-k}$, where k is the number of successes. We are given $n = 5$ and a dataset of $N = 10$ observations: $\{x_1, x_2, \ldots, x_{10}\} = \{3, 4, 2, 5, 4, 3, 2, 4, 5, 3\}$. The likelihood function for this dataset, given p, is the product of the PMFs for each observation.

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35 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

Consider a medical diagnostic test for a rare disease. The prevalence of the disease in the general population is $P(D) = 0.001$. The test has a sensitivity of $P(+|D) = 0.99$ (the probability of a positive test given the person has the disease) and a specificity of $P(-|egD) = 0.98$ (the probability of a negative test given the person does not have the disease). If a randomly selected individual tests positive, what is the probability that they actually have the disease, given that we are interested in the positive predictive value (PPV) of the test, and the prior probability of having the disease is adjusted by a factor f such that the new prior is $P'(D) = f \cdot P(D)$? For this specific scenario, suppose $f = 1.5$ and we want to find $P'(D|+)$.

a. $\frac{0.99 \cdot 0.0015}{0.99 \cdot 0.0015 + 0.02 \cdot (1 - 0.0015)}$ ✓

b. $\frac{0.99 \cdot 0.0015}{0.99 \cdot 0.0015 + 0.02 \cdot 0.0015}$

c. $\frac{0.99 \cdot 1.5}{0.99 \cdot 1.5 + 0.02 \cdot (1 - 1.5)}$

d. $\frac{0.99 \cdot 0.0015}{0.99 \cdot 0.0015 + 0.98 \cdot (1 - 0.0015)}$

Let D be the event that an individual has the disease, and + be the event that the test is positive. We are given: - Prior probability of disease: $P(D) = 0.001$ - Sensitivity: $P(+|D) = 0.99$ - Specificity: $P(-|egD) = 0.98$ From specificity, we can find the false positive rate: $P(+|egD) = 1 - P(-|egD) = 1 - 0.98 = 0.02$. We are considering a scenario where the prior probability is adjusted by a factor $f = 1.5$. The new prior probability of having the disease is.

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36 [Intermediate] • Target: Vectors & Embeddings in AI Practice Exercises

Consider a univariate regression model where the predicted output $\hat{y}$ is given by $\hat{y} = ax + b$. The loss function is the Mean Squared Error (MSE), defined as $L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. $For a single data point (x_1, y_1) = (2, 5), the loss is L = (5 - (2a + b))^2. The gradient of the loss function with respect to a is \frac{\partial L}{\partial a} = -2(5 - (2a + b)) \cdot 2. If during gradient descent, with a learning rate of \eta = 0.1, the parameters a and b are updated from a_{old} = 1 and b_{old} = 1, to a_{new} = 0.8 and b_{new} = 0.8, what is the value of b that would have resulted in \frac{\partial L}{\partial b} = 0 for this single data point, assuming a=1 and b=1 were the values * before * the gradient calculation that led to the update to a_{new}=0.8, b_{new}=0.8$?

a. 3

b. 4

c. 5 ✓

d. 6

The loss function for a single data point is $L = (y_1 - (ax_1 + b))^2$. $Given (x_1, y_1) = (2, 5), so L = (5 - (2a + b))^2. The gradient of the loss with respect to b is \frac{\partial L}{\partial b} = 2(5 - (2a + b)) \cdot (-1) = -2(5 - 2a - b). We are asked to find the value of b for which \frac{\partial L}{\partial b} = 0, assuming a=1 and b=1 were the parameters * before * the update step that led to a_{new}=0.8, b_{new}=0.8. This means we evaluate the gradient at a=1 and the * unknown * value of b. $Set.

37 [Intermediate] • Target: Vectors & Embeddings in AI Practice Exercises

A dataset consists of observations $x_1, x_2, \ldots, x_n$ that are assumed to be drawn independently from a Bernoulli distribution with parameter p. The probability mass function is given by $P(X = k; p) = p^k(1-p)^{1-k}$ for $k \in \{0, 1\}$. The following bar chart shows the frequency distribution of these observations, where $n = 50$. { "type": "bar", "labels": ["0", "1"], "data": [20, 30], "title": "Frequency Distribution of Bernoulli Observations" } Using the method of Maximum Likelihood Estimation (MLE), determine the estimated value of p for this distribution.

a. $\frac{2}{5}$

b. $\frac{3}{5}$ ✓

c. $\frac{1}{2}$

d. $\frac{3}{2}$

The Bernoulli distribution has the probability mass function $P(X = k; p) = p^k(1-p)^{1-k}$. The likelihood function for n independent observations is $L(p; x_1, \ldots, x_n) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}$. This can be rewritten as $L(p) = p^{\sum x_i}(1-p)^{n-\sum x_i}$. The log-likelihood function is $\ln L(p) = (\sum x_i)\ln p + (n - \sum x_i)\ln(1-p)$. To find the maximum likelihood estimate (MLE) of p, we take the derivative of the log-likelihood with respect to p and set it to zero.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A dataset consists of three independent observations X_1, X_2, X_3 drawn from a Poisson distribution with parameter λ. The observed values are $x_1 = 3, x_2 = 5, x_3 = 4$. We want to find the Maximum Likelihood Estimate (MLE) for λ. The log-likelihood function for a Poisson distribution is given by $\ell(\lambda) = \sum_{i=1}^{n} (x_i \log(\lambda) - \log(x_i!) - \lambda)$. Part 1: Derive the expression for the derivative of the log-likelihood function with respect to λ. Part 2: Set the derivative to zero to find the MLE of λ.

a. $\hat{\lambda} = \frac{3+5+4}{3} = 4$ ✓

b. $\hat{\lambda} = \frac{3+5+4}{3 \times e}$

c. $\hat{\lambda} = \frac{\sum x_i}{\sum \log(x_i!)}$

d. $\hat{\lambda} = \frac{1}{3}$

The log-likelihood function for $n = 3$ observations is $\ell(\lambda) = (x_1 \log(\lambda) - \log(x_1!) - \lambda) + (x_2 \log(\lambda) - \log(x_2!) - \lambda) + (x_3 \log(\lambda) - \log(x_3!) - \lambda)$. Part 1: The derivative of the log-likelihood function with respect to λ is: $\frac{d\ell}{d\lambda} = \sum_{i=1}^{3} \left(\frac{x_i}{\lambda} - 1\right) = \left(\frac{x_1}{\lambda} - 1\right) + \left(\frac{x_2}{\lambda} - 1\right) + \left(\frac{x_3}{\lambda} - 1\right)$. Part 2: To find the MLE.

39 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

A medical diagnostic test for a rare disease is designed to be highly sensitive and specific. The disease affects 1 in every 1000 individuals in the population. The test has a sensitivity of 99% (meaning it correctly identifies 99% of individuals who have the disease) and a specificity of 95% (meaning it correctly identifies 95% of individuals who do not have the disease). If a randomly selected individual from the population tests positive, what is the probability that they actually have the disease?

a. **Approximately** 0.019 ✓

b. Approximately 0.99

c. Approximately 0.05

d. Approximately 0.95

Let D be the event that an individual has the disease, and P be the event that the individual tests positive. We are given the following probabilities: - Prior probability of having the disease: $P(D) = \frac{1}{1000} = 0.001$ - Prior probability of not having the disease: $P(D^c) = 1 - P(D) = 1 - 0.001 = 0.999$ - Sensitivity (true positive rate): $P(P|D) = 0.99$ - Specificity (true negative rate): $P(P^c|D^c) = 0.95$ From specificity, we can derive the false positive rate.

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40 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

A 2D linear transformation T is defined by the matrix $M = (3, 1; 1, 3)$. Consider a triangle with vertices $A = (1, 0)$, $B = (0, 1)$, and $C = (1, 1)$. This triangle is transformed by T to a new triangle with vertices A', B', and C'. The area of the transformed triangle $A'B'C'$ is then passed through a ReLU activation function, $f(x) = \max(0, x)$. What is the final output value?

a. 8

b. 4 ✓

c. 0

d. 16

The area of the original triangle with vertices $A = (1, 0)$, $B = (0, 1)$, and $C = (1, 1)$ can be calculated. We can use the determinant formula for the area of a triangle given its vertices (x_1, y_1), (x_2, y_2), (x_3, y_3): Area = $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$. Plugging in the coordinates: Area = $\frac{1}{2}|1(1 - 1) + 0(1 - 0) + 1(0 - 1)| = \frac{1}{2}|0 + 0 - 1| = \frac{1}{2}$. A linear transformation represented by a matrix M scales the area of a 2D shape by the abs.

41 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider a dataset of n data points, each residing in $\mathbb{R}^D$. We are performing Principal Component Analysis (PCA) to reduce the dimensionality. The first three principal components capture 70% of the total variance, the first six capture 85%, and the first ten capture 92%. You are tasked with selecting a subset of principal components for a downstream machine learning model. If the model requires that at least 90% of the original variance be retained, and you want to use the minimum possible number of principal components, what is the dimensionality of the transformed data?

a. 3

b. 6

c. **10** ✓

d. D

The problem requires us to find the minimum number of principal components such that the cumulative explained variance is at least 90%. The provided information states: - The first 3 principal components capture 70% of the variance. - The first 6 principal components capture 85% of the variance. - The first 10 principal components capture 92% of the variance. We need to find the smallest k such that the cumulative variance captured by the first k principal components is $\geq 90\%$. Checking the giv.

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42 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider a dataset of observations $x_1, x_2, \ldots, x_n$ assumed to be drawn independently from an exponential distribution with probability density function (PDF) $f(x;\lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$ and $\lambda > 0$. The likelihood function for this dataset is given by $L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i}$. To find the Maximum Likelihood Estimator (MLE) for λ, we first consider the log-likelihood function $\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i$. If the observed data points are $\{1.2, 2.5, 0.8, 3.1, 1.5 \backslash)$, and $n = 5$, determine the MLE of λ.

a. $\frac{1}{1.62}$

b. $\frac{5}{9.1}$ ✓

c. $\frac{1}{5} \cdot \frac{9.1}{\ln(5)}$

d. $\frac{1}{\bar{x}}$

The log-likelihood function is $\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i$. To find the MLE, we differentiate $\ln L(\lambda)$ with respect to λ and set the derivative to zero: $\frac{d}{d\lambda} \ln L(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i$ Setting this to zero: $\frac{n}{\lambda_{MLE}} - \sum_{i=1}^{n} x_i = 0$ $\frac{n}{\lambda_{MLE}} = \sum_{i=1}^{n} x_i$ $\lambda_{MLE} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\frac{1}{n}\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}}$ The observed data point.

43 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider a binary classification problem where the decision boundary is defined by the logistic function $p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$, which represents the probability of a data point $\mathbf{x} = (x_1, x_2)$ belonging to class 1. The estimated model parameters are $\beta_0 = -3$, $\beta_1 = 1$, and $\beta_2 = -2$. We are interested in the geometric shape of the classification boundary. The classification boundary occurs where $p(x) = 0.5$. Find the equation of this boundary in the x_1x_2-plane.

a. $x_2 = \frac{1}{2}x_1 - \frac{3}{2}$ ✓

b. $x_2 = \frac{1}{2}x_1 + \frac{3}{2}$

c. $x_2 = 2x_1 - 3$

d. $x_2 = 2x_1 + 3$

The logistic regression model gives the probability of belonging to class 1 as $p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$. The classification boundary is the set of points where $p(x) = 0.5$. Setting $p(x) = 0.5$, we have: $0.5 = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$ This implies: $1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)} = 2$ Therefore: $e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)} = 1$ Taking the natural logarithm of both sides.

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44 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

A medical test for a rare disease has a 99\% accuracy rate in detecting the disease when it is present (True Positive Rate) and a 95\% accuracy rate in identifying individuals without the disease (True Negative Rate). If the prevalence of the disease in the population is 0.5\%, what is the probability that a person who tests positive actually has the disease?

a. Approximately 8.7\%

b. **Approximately 9.1\% ✓**

c. Approximately 1.0\%

d. Approximately 0.5\%

Let D be the event that a person has the disease, and P be the event that the person tests positive. We are given: P(P|D) = 0.99 (True Positive Rate) P(not P|not D) = 0.95 (True Negative Rate), which means P(P|not D) = 1 - 0.95 = 0.05 (False Positive Rate) P(D) = 0.005 (Prevalence of the disease) From P(D), we know P(not D) = 1 - P(D) = 1 - 0.005 = 0.995. We want to find P(D|P), the probability that a person has the disease given a positive test result. Using Bayes' Theorem: P(D|P) = [P(P|D) * P(D)] / P(P) To find.

45 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A dataset consists of $n = 100$ independent and identically distributed (i.i.d.) observations $x_1, x_2, \ldots, x_{100}$. These observations are drawn from a Bernoulli distribution with parameter p. The probability mass function (PMF) of a Bernoulli distribution is given by $P(X = x; p) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$. The likelihood function $L(p)$ for this dataset is the product of the PMFs of the individual observations. If we observe $k = 60$ successes (i.e., $x_i = 1$) and $n - k = 40$ failures (i.e., $x_i = 0$), what is the likelihood function $L(p)$ for p?

a. $p^{60}(1-p)^{40}$ ✓

b. $p^{100}(1-p)^{100}$

c. $60p(1-p)^{40}$

d. $40p^{60}(1-p)$

The likelihood function $L(p)$ is the product of the probabilities of observing each data point given the parameter p. For i.i.d. Bernoulli trials, $L(p) = \prod_{i=1}^{n} P(X = x_i; p)$. Given $n = 100$ observations, with $k = 60$ successes (where $x_i = 1$) and $n - k = 40$ failures (where $x_i = 0$), the likelihood function can be written as: $L(p) = \left(\prod_{i:x_i=1} p^1(1-p)^{1-1}\right) \times \left(\prod_{i:x_i=0} p^0(1-p)^{1-0}\right)$.

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